

Operational Shadows of Hilbert-Space Probabilities

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At one frozen setting, the probabilities observed in a sharp quantum context are indistinguishable, as detector-click statistics, from ordinary probabilities on the atoms of a classical partition. But an actual analyzer usually comes with a calibrated knob: a tangible handle on the apparatus. If the measurement configuration is co-varied continuously through this physical parameter, the operational object is no longer one point of a simplex but a response curve. Classical linear responses, Malus-type Hilbert-space responses, softmax links, non-homomorphic parameter transcriptions, and discontinuous threshold limits are different maps from settings to probabilities. Continuity, calibration, and preservation of the physical composition law are then part of the experimental meaning of the knob. Such comparisons distinguish specified, calibrated response models; by themselves they do not constitute a classical-versus-quantum impossibility theorem. The static operational coincidence can also persist for two intertwined contexts: if the common outcomes receive the same probabilities, the remaining masses can always be coupled by a classical joint distribution. Genuine multi-context nonclassicality begins when a family of local shadows cannot be glued into one non-negative global distribution or one simplex factorization. Farkas’ lemma gives the exact alternative: either the classical extension exists, or a separating linear inequality certifies its impossibility.

I. INTRODUCTION

A hand held before a lamp casts a flat shadow on a wall. Many different hands, postures, and objects can cast the same silhouette. If the wall is the only thing one is allowed to inspect, the shadow is all one has; but it should not be mistaken for the object that produced it. This is the old warning of Plato’s cave: the prisoners take the passing shadows for reality. The image contains a second warning that is just as relevant here. The one who turns around and returns with an account of the objects and the light behind the shadows is not automatically believed; he may appear blinded or confused, and the cave’s inhabitants may punish the attempt to free them from their own evidence [1, Book VII, 514a–517a]. Thus the metaphor cuts both ways. A shadow should not be reified as the thing itself, but neither is the hidden machinery behind it given by the shadow alone. To speak of an ontology casting the shadow is already to add a reconstruction, a model of what lies off the wall.

Something similar happens in a single sharp measurement. A system is sent into a detector bank, one detector clicks, and after many repetitions one has a list of relative frequencies. This list is a point of a probability simplex. Looking only at that list, one cannot tell whether it was generated by a classical partition, by squared Hilbert-space amplitudes, or by a more exotic admissible weight rule. The generating geometries are different, but a single frozen setting can flatten them into the same operational shadow.

The missing character is often the knob: a tangible handle on the apparatus. A polarizer rotates, a Stern–Gerlach magnet tilts, a phase is advanced, or, in a more

abstract response model, a temperature or score is dialed. In the rotational examples, turning the knob is itself part of the experiment: equal increments can be calibrated, small changes can be made continuously, and successive transformations compose. The knob is therefore not merely a coordinate on an interval; it carries a physical action. Once this variation is retained, the shadow is no longer a single point but a trajectory through the simplex. A specified classical response model may hit any one point of a Hilbert-space response curve, but it need not reproduce the whole curve with the same symmetry-preserving knob. This is the intuition behind the “parameter cheats”: they work by deforming the scale on which the knob is read, thereby changing the physical action represented by equal turns of the apparatus.

This article separates four levels. One static context gives an operational shadow. A calibrated sweep compares response laws within a specified model class. Two compatible context distributions always admit a classical coupling. Only a sufficiently constrained web of contexts can fail to extend to one global nonnegative distribution, in which case Farkas’ lemma yields a separating inequality. The response-law discussion is methodological: a single calibrated curve does not, without additional covariance, locality, or ontic-response assumptions, certify Hilbert-space probability or exclude arbitrary classical stochastic models.

A continuously swept apparatus can indeed be represented formally as a continuum of mutually incompatible quantum contexts: each fixed setting determines a different maximal projective measurement. That observation is not, however, the main use of the knob in this article. The sweep is treated first as an experimentally calibrated response measurement—a map from a physical control variable to click probabilities—rather than as a quantum-logical pasting problem over a continuum of contexts. The group action and continuity belong to the kinemat-

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ics of the apparatus; they do not by themselves supply a nonclassicality witness or an orthomodular-algebraic obstruction. Such an obstruction arises only when one asks whether the probabilities from a specified family of contexts can be glued into one global nonnegative model.

II. ONE STATIC CONTEXT

Let $C = \{P_1, \dots, P_d\}$ be a maximal rank-one projective measurement on a d -dimensional Hilbert space. In a run of the experiment only one detector clicks, and the operational data associated with this single context are the frequencies

$$p_i \geq 0, \quad \sum_{i=1}^d p_i = 1. \quad (1)$$

They are the same kind of object as a probability distribution on the atoms of a classical partition. The phrase “same operational shadow”¹ will mean this: the observable click statistics in the context are identical, even if the underlying state spaces used to generate them are different.

The difference between the two theories is already present in the generating maps. A classical state on a d -atom partition is a point of the simplex Δ_{d-1} , with pure states at the vertices and mixtures represented affinely. A pure quantum state is a ray in complex projective space, $[\psi] \in \mathbb{CP}^{d-1}$, and Born’s rule gives

$$s_C([\psi]) = (\langle \psi | P_1 | \psi \rangle, \dots, \langle \psi | P_d | \psi \rangle). \quad (2)$$

In an eigenbasis $P_i = |e_i\rangle\langle e_i|$, this is the squared projection map $p_i = |\langle e_i | \psi \rangle|^2$. Thus the same probability simplex is reached through a different geometry: affine weights in the classical case, squared Hilbert-space amplitudes in the quantum case.

Proposition 1. *For one maximal rank-one context C , the pure-state shadow map $s_C : \mathbb{CP}^{d-1} \rightarrow \Delta_{d-1}$ is onto.*

Proof. Given $p = (p_1, \dots, p_d) \in \Delta_{d-1}$, choose arbitrary phases ϕ_i and set

$$|\psi\rangle = \sum_{i=1}^d \sqrt{p_i} e^{i\phi_i} |e_i\rangle. \quad (3)$$

Then $\langle \psi | P_i | \psi \rangle = p_i$. The phases, modulo a global phase, are invisible to this context. Hence a single context sees the full ordinary probability simplex and nothing beyond it. $\square$

¹ This use of “shadow” is the Plato-cave metaphor used here, not the randomized measurement technique known in quantum information as “classical shadows”.

This proposition is deliberately modest. It does not say that the classical and quantum state spaces are isomorphic. It says only that after forgetting everything except one detector bank’s click frequencies, both theories cast the same operational shadow. The phrase “forgetting” is essential: it freezes the apparatus setting and discards the way in which the probabilities change when the apparatus is moved.

III. CALIBRATED RESPONSE LAWS

In the laboratory the setting is rarely just an abstract label. A polarizer is rotated, a Stern–Gerlach magnet is tilted, a phase is advanced, or a score parameter is changed. The detector bank may still be the same operational device, but the measurement configuration is co-varied with a physical parameter. Once this knob is retained, the observed data are not merely $p \in \Delta_{d-1}$, but the curve swept out by p as the knob is turned. If K denotes the setting space of the knob, the relevant object is the response function

$$R_C : K \longrightarrow \Delta_{d-1}, \quad k \longmapsto (p_1(k), \dots, p_d(k)). \quad (4)$$

A classical partition can hit any one point of this curve. That is not the same as reproducing the curve with the same independently calibrated physical action.

A calibrated knob is not just an arbitrary coordinate on an interval. More generally, its setting space K carries the action of a physically implemented transformation group G , and is often a homogeneous space G/H . An admissible change of coordinates should preserve this action, or at least intertwine it with the corresponding physical action on the apparatus. Continuity is part of this structure, but it is not the whole structure: a continuous deformation of the coordinate can still fail to preserve the action.

For a one-parameter planar rotation, the composable knob variable should be a signed angle $\varphi \in \mathbb{R}/2\pi\mathbb{Z}$, or the corresponding quotient appropriate to unoriented axes. The scalar $\theta \in [0, \pi]$ used below is an unsigned relative separation derived from that action. It is suitable as the argument of an isotropic response curve, but it does not itself obey a global addition law. For general analyzer orientations the acting group is $SO(3)$, and the scalar separation likewise should not be identified with a group element. A harmless reparameterization must therefore intertwine the physical group action, not merely relabel the scalar argument of a plotted curve. Such calibration can be checked independently of the output probabilities by composing transformations, verifying the appropriate period, or using a phase reference. Equal increments, composed rotations, and continuous sweeps are operationally implementable and comparable.

One may object that, once the knob is turned, one has already left the original context and entered a many-context situation. Formally this can be a useful description: each fixed angle may be represented by its own

maximal context. But that is not the emphasis here. A knob sweep is not primarily a finite list of unrelated configurations to be glued by a marginal-extension test. It is a functional response to an argument, such as $\theta \mapsto p(\theta)$, where the argument has a physical scale, continuity, and composition or symmetry law. The question is therefore not only which static contexts are present, but how the probabilities vary when the apparatus is continuously moved.

Within a specified model class, this is already enough to distinguish response laws along a single calibrated sweep. For example, a deterministic shared-direction model with a uniformly distributed hidden orientation and a threshold readout gives a linear equal-outcome response, whereas a spin-1/2 singlet gives the curved Malus-type response. This comparison does not exclude all classical stochastic response models for one curve; locality and global-consistency restrictions enter only when several settings must be represented together. For these binary examples write $R(\theta) = P^=(\theta)$. On $0 \leq \theta \leq \pi$ one may write

$$\begin{aligned} P_{\text{cl}}^=(\theta) &= \frac{\theta}{\pi}, \\ P_{\text{qm}}^=(\theta) &= \sin^2 \frac{\theta}{2}, \\ P_{\text{s}}^=(\theta) &= H\left(\frac{2\theta}{\pi} - 1\right), \end{aligned} \quad (5)$$

where $H(0) = 1/2$ fixes the convention at the threshold. The last line is a discontinuous toy response, sometimes used as a stronger-than-quantum limit in a multi-setting construction [2]; the curve by itself is not a no-signalling multi-setting model. At a single value of θ , each line gives only an ordinary two-outcome probability distribution. As θ is co-varied, however, the variational response is completely different: linear, smoothly curved, or step-like.

The 2001 “parameter cheat” construction makes this point particularly transparent. If one introduces a new parameter δ by $\theta = \pi \sin^2(\delta/2)$, then the classical expression $P_{\text{cl}}^=(\theta) = \theta/\pi$ can be made to look quantum:

$$P_{\text{cl}}^=(\theta(\delta)) = \sin^2 \frac{\delta}{2}.$$

Conversely, the quantum curve can be made to look linear by displaying it against the parameter $\eta = \pi \sin^2(\theta/2)$. Pointwise, the trick works. But the price is that the new parameter is no longer the same physical knob. The transcription is not a homomorphism, or more generally an equivariant map, for the calibrated rotation action [3]. It therefore changes the calibration through which the apparatus is varied rather than merely relabeling probabilities.

A. Softmax links and inverse knobs

Softmax belongs to the same discussion, although its natural knob is often a score, utility, or inverse-temperature parameter rather than a literal angle. It is included here only as an abstract response law into a simplex, not as a physical analogue of Born’s quadratic amplitude rule. Given scores $u_C(a)$ and a strictly positive, usually monotone, link g , it forms

$$P_{g,C}(a) = \frac{g(u_C(a))}{\sum_{b \in C} g(u_C(b))}. \quad (6)$$

For $g_\beta(u) = \exp(\beta u)$, the parameter β interpolates between an almost uniform response at small β and a winner-take-all threshold as $\beta \rightarrow \infty$. This is useful as a link-function deformation, not as a claim that the exponential family has the same physical origin as Born’s rule.

Changing g , or changing the score scale, changes the response sensitivity of the model. For a single context this again guarantees only local normalization; it does not by itself specify which score scale is physically proper. If the same atom occurs in several contexts, admissibility adds a separate requirement: the response must be single-valued on that atom. Thus a softmax fit, a Malus curve, a linear classical curve, and a Heaviside step are not competing physical explanations. They are different response functions tied to different assumptions about what is being varied and which structure that variation is required to preserve.

For a single monotone response curve this observation can be made completely explicit. Let $R_A(\theta)$ and $R_B(\theta)$ be two response functions, written here as equal-outcome probabilities. Equivalently, one may use the correlation $E(\theta) = 2R(\theta) - 1$. If R_A is invertible on the range of R_B , then

$$T_{A \leftarrow B}(\theta) = R_A^{-1}(R_B(\theta)) \quad (7)$$

is the transcribed knob: feeding $T_{A \leftarrow B}(\theta)$ into response law A makes it reproduce response law B as a function of the displayed parameter θ . The injectivity assumption is substantive. Without a single-valued inverse one obtains at best a set-valued transcription or a limiting prescription, as in a threshold response.

For the classical linear response $R_{\text{cl}}(\theta) = \theta/\pi$ and the quantum singlet response $R_{\text{qm}}(\theta) = \sin^2(\theta/2)$, this gives

$$T_{\text{cl} \leftarrow \text{qm}}(\theta) = \pi \sin^2 \frac{\theta}{2}, \quad T_{\text{qm} \leftarrow \text{cl}}(\theta) = 2 \arcsin \sqrt{\frac{\theta}{\pi}}. \quad (8)$$

These are precisely the parameter cheats: the first makes a classical linear response look quantum; the second makes the quantum response look classical.

For a centered binary softmax response with parameter $\beta > 0$, define

$$P_{\sigma,\beta}^=(\theta) \equiv R_{\sigma,\beta}(\theta) = \{1 + \exp[-\beta(2\theta/\pi - 1)]\}^{-1}. \quad (9)$$

This is the softmax counterpart of the response laws in Eq. (5). As $\beta \downarrow 0$ it approaches the uniform (constant-in- θ) binary response $P^=(\theta) = 1/2$; as $\beta \rightarrow \infty$ it approaches the Heaviside threshold response (with convention $H(0) = 1/2$).

The corresponding inverse knobs are

$$T_{\text{cl} \leftarrow \sigma}(\theta) = \pi R_{\sigma, \beta}(\theta), \quad (10)$$

$$T_{\sigma \leftarrow \text{cl}}(\theta) = \frac{\pi}{2} \left[1 + \frac{1}{\beta} \log \frac{\theta/\pi}{1 - \theta/\pi} \right], \quad (11)$$

where the second formula is finite only for $0 < \theta/\pi < 1$. For an input knob restricted to $[0, \pi]$, the inverse $T_{\sigma \leftarrow \text{cl}}(\theta)$ lies in that physical interval precisely when

$$\frac{1}{1 + e^\beta} \leq \frac{\theta}{\pi} \leq \frac{1}{1 + e^{-\beta}}. \quad (12)$$

Outside this range the inverse is still a formal transcription, but not an admissible setting of the bounded softmax knob. Similarly,

$$T_{\text{qm} \leftarrow \sigma}(\theta) = 2 \arcsin \sqrt{R_{\sigma, \beta}(\theta)},$$

and $T_{\sigma \leftarrow \text{qm}}$ is obtained by inserting $\sin^2(\theta/2)$ into the logit expression in Eq. (11). The Heaviside response $R_H(\theta) = H(2\theta/\pi - 1)$ can be transcribed into the classical knob as $T_{\text{cl} \leftarrow H}(\theta) = \pi R_H(\theta)$, but it has no ordinary inverse: an interval of angles is collapsed to one of two values. Its inverse is set-valued, or else appears only as a limiting threshold.

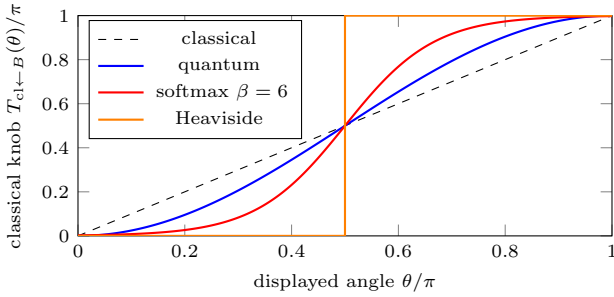


FIG. 1. Classical inverse knobs for several target response laws. Since $R_{\text{cl}}(\tau) = \tau/\pi$, the transcribed classical knob is simply $T_{\text{cl} \leftarrow B}(\theta)/\pi = R_B(\theta)$. The curves show how a linear classical response must be read on a generally non-symmetry-preserving scale in order to mimic a quantum, softmax, or Heaviside response as a function of the displayed angle.

This single-curve trick should not be confused with a solution of a Bell or Boole problem. In a Bell–Clauser–Horne–Shimony–Holt (CHSH) experiment one does not merely fit one function $E(\theta)$. One must assign probabilities to several settings at once, such as a, b, a', b' , with one common underlying distribution over predetermined answers. A separate inverse knob can make each pairwise correlation look right, but those pairwise deformations need not come from one action-equivariant calibration.

For a signed planar parameter φ , the transcribed map generally fails to satisfy

$$T(\varphi_1 + \varphi_2) = T(\varphi_1) + T(\varphi_2);$$

for general analyzer orientations, the corresponding failure is nonintertwining of the $SO(3)$ action. Thus the deformed pairwise settings need not be compatible with one geometry of four settings. The marginal-extension problem then reappears, and Farkas’ lemma supplies the relevant obstruction.

Elastic-band examples make this warning concrete. Aerts’ spin machine and its extended-Bloch descendants use a hidden break point, or more generally a hidden measurement interaction, to generate Born-like responses from an ordinary macroscopic mechanism [6–8]. The rubber-band analysis of context communication stresses the complementary limitation: recovering a cosine-shaped pair correlation in one adapted arrangement is not yet the same as a Bell-local simulation of an entire CHSH family [9]. If the band or shared resource has to be realigned when the selected pair $(a, b), (a, b'), (a', b), (a', b')$ changes, then the construction has acquired setting dependence or context communication. The shadow can be made to look quantum after breakfast, lunch, teatime, and supper, but the four shadows no longer arise from one setting-independent joint valuation.

In Sven Aerts’ Popescu–Rohrlich-box rubber-band construction, the same point appears in a more extreme form: the resource is one extended object whose breaking co-creates the two outcomes [10]. These models are relevant here not as refutations of Boole–Bell, but as physical parables of where the response-function program stops. A single curve can be reparameterized, whereas a Bell web requires one common nonadaptive geometry or else a setting-dependent, nonlocal, or nonseparable resource.

IV. TWO CONTEXTS AND GLOBAL EXTENSIONS

The same logic explains why two contexts, even when they share outcomes, are not yet enough to force a probability-level obstruction. Let

$$A = S \cup X, \quad B = S \cup Y \quad (13)$$

be two Boolean blocks, with shared atoms S , atoms X belonging only to A , and atoms Y belonging only to B . Suppose context-wise probabilities p^A and p^B are normalized, nonnegative, and agree on the overlap:

$$p_s^A = p_s^B =: r_s \quad (s \in S). \quad (14)$$

Let $m = 1 - \sum_{s \in S} r_s$. Then

$$\sum_{x \in X} p_x^A = m = \sum_{y \in Y} p_y^B. \quad (15)$$

If $m > 0$, choose, for example,

$$w_{xy} = \frac{p_x^A p_y^B}{m}; \quad (16)$$

if $m = 0$, set all $w_{xy} = 0$. Now define a classical sample space with one point λ_s of weight r_s for each shared atom s , and one point λ_{xy} of weight w_{xy} for each pair $x \in X, y \in Y$. At λ_s , the shared atom s is the true answer in both contexts. At λ_{xy} , the answer to A is x and the answer to B is y . Reading the A -coordinate gives p^A , and reading the B -coordinate gives p^B .

Proposition 2. *Every pair of normalized context distributions satisfying Eq. (14) has a classical joint model over deterministic answers for the two blocks.*

This is just the elementary coupling theorem for two marginal distributions, with the shared outcomes treated as already coupled. Equivalently, after the common mass has been fixed, each residual cell obeys the finite Fréchet–Hoeffding bounds

$$\max(0, p_x^A + p_y^B - m) \leq w_{xy} \leq \min(p_x^A, p_y^B), \quad (17)$$

and Eq. (16) is just one point in this coupling polytope. Thus two-context intertwining has no nontrivial marginal obstruction at the level of static weights. The response functions realizing those weights may still be very different: continuous or discontinuous, affine or Malus-type, softmax-shaped or boundary-valued. The obstruction requires a sufficiently constrained web of local shadows, typically a cycle or a Kochen–Specker-type pasting, so that all pairwise or context-wise agreements cannot be realized by one global nonnegative distribution.

A. Farkas alternative

The extension problem can be written in the form

$$Mq = b, \quad q \geq 0, \quad (18)$$

where q is a nonnegative distribution over deterministic global assignments and b collects the observed context probabilities, including the relevant normalization constraints. Farkas’ lemma says that exactly one of the following alternatives holds:

$$\exists q \geq 0 \text{ such that } Mq = b, \quad (19)$$

$$\exists y \text{ such that } y^T M \geq 0 \text{ and } y^T b < 0. \quad (20)$$

The second line is a separating inequality. Garg and Mermin used precisely this logic to formulate the question of whether specified pair distributions can arise as marginals of compatible higher-order distributions [11]. Fine’s theorem for Bell inequalities is a celebrated special case of the same joint-distribution viewpoint [12]; Pitowsky’s correlation polytopes place the construction in convex-geometric form [13].

Equation (18) is also the finite-dimensional skeleton of simplex-embeddability tests. A finite prepare-and-measure fragment is classical when its probability data admit a factorization through a simplex. Once the relevant extremal classical response assignments and linear operational equivalences have been enumerated, the unknown convex weights obey a feasibility system of the form (18). Infeasibility then has a Farkas dual certificate. Recent generalized-probabilistic simplex criteria and their linear-programming implementations make this operational version explicit [4, 5].

V. THE PENTAGON: CLASSICAL, QUANTUM, AND ADMISSIBLE BODIES

For a fixed exclusivity structure there are three natural nested convex sets: a classical body, a Hilbert-space body, and a larger admissible-weight body. The classical body is generated by deterministic two-valued states:

$$\mathcal{P}_{\text{cl}} = \text{conv}\{\text{admissible 0/1 assignments}\}. \quad (21)$$

For graphs this is the stable-set polytope, denoted $\text{STAB}(G)$, possibly with additional normalization equations for complete contexts.

One standard realization of the corresponding quantum body is obtained from squared projections of a state onto an orthogonal representation:

$$p_v = |\langle \psi | v \rangle|^2, \quad (22)$$

or by the corresponding density-operator formula. In graph language this is related to Lovász-type theta bodies [14–16]. The still larger admissible-weight body, often described by no-disturbance constraints in contextuality scenarios, is obtained by retaining only nonnegativity, context-wise normalization, additivity on exclusive events, and equality of the values assigned to the same atom in different contexts.

For one clique, these three bodies have the same operational shadow: the simplex. For two cliques pasted along a common clique, the explicit coupling establishes only that there is no classical extension obstruction for the compatible context marginals. It does not imply that every such pair is realizable by a fixed pair of quantum projective contexts.

Cycles change the situation. The pentagon is the minimal cyclic illustration. Let

$$C_i = \{a_i, a_{i+1}, x_i\}, \quad i = 1, \dots, 5,$$

with indices modulo 5. The cyclic atoms a_i are the intertwining observables, and the atoms x_i are context-specific. The assignment

$$p(a_i) = \frac{1}{2}, \quad p(x_i) = 0 \quad (i = 1, \dots, 5) \quad (23)$$

obeys completeness in every context,

$$p(a_i) + p(a_{i+1}) + p(x_i) = 1,$$

is additive on mutually exclusive events inside each Boolean block, and is atom-consistent: the same intertwining atom has the same value wherever it occurs. This is deliberately weaker than noncontextual hidden-variable embeddability in a classical simplex, and it is not Spekkens-style operational noncontextuality. In particular, it is not a convex mixture of deterministic two-valued states.

The classical noncontextual bound on the five cyclic events is

$$\sum_{i=1}^5 p_i \leq 2, \quad (24)$$

which is the Klyachko–Can–Binicioğlu–Shumovsky (KCBS) pentagon inequality [17–19]. By contrast, a faithful quantum orthogonal representation reaches

$$\sum_{i=1}^5 p_i = \sqrt{5}. \quad (25)$$

This is the Lovász number of the pentagon exclusivity graph [16, 20].

The half-weight (23) gives $\sum_i p(a_i) = 5/2$, so it is admissible and atom-consistent, yet lies beyond both the classical bound and the usual KCBS/Lovász quantum value. It is therefore an explicit admissible, atom-consistent, but post-quantum point in the gap between the classical and Hilbert-space bodies. In the language used here, the Hilbert-space squared-projection body escapes the classical simplex body only after enough context-wise shadows have been required to glue together; the exotic half-weight escapes still farther inside the larger admissible/no-disturbance polytope. It is best read as a boundary object of the admissible theory, not as a Hilbert-space-realizable quantum state.

The KCBS inequality can be written as an exact Farkas certificate, but the all-ones row defining the inequality is not by itself in the sign convention of Eq. (20). Let

$$c^T b = \sum_{i=1}^5 p(a_i),$$

and augment the system $Mq = b$ by a normalization row $n^T b = 1$, or, equivalently, use the corresponding linear combination of the context-normalization rows. Every deterministic assignment obeys

$$c^T M \leq 2n^T M.$$

Hence the Farkas vector $y = 2n - c$ satisfies

$$y^T M \geq 0, \quad y^T b = 2 - \frac{5}{2} = -\frac{1}{2} < 0. \quad (26)$$

This is precisely the dual certificate for the nonexistence of a nonnegative convex combination of deterministic valuations realizing $b = (1/2, \dots, 1/2)$.

The half-weight can be approached inside the admissible polytope by the one-parameter path

$$p_r(a_i) = \frac{1}{2+r}, \quad p_r(x_i) = \frac{r}{2+r}, \quad r \geq 0. \quad (27)$$

The uniform point occurs at $r = 1$, whereas the exotic half-weight is the boundary limit $r \rightarrow 0$.

This path also admits a generalized-softmax reading. If the two cyclic atoms in each context receive a common score u_0 , the context-specific atom receives a score t , and

$$r = \frac{g(t)}{g(u_0)},$$

then Eq. (6) gives precisely Eq. (27). For a strictly positive exponential link, $r = 0$ is reached only as a limiting response rather than at finite scores; a Heaviside-type link can instead reach the boundary discontinuously. This algebraic representation is only an admissible interpolation. It does not claim that every point on the path is generated by one calibrated physical mechanism. Rather, it reinforces the distinction between static consistency, classical embeddability, Hilbert-space realizability, and a calibrated dynamical response law.

VI. FREEZE THE CONTEXT AND CO-VARY THE PREPARATION

One may equally well freeze the context and co-vary the preparation: a half-wave plate before a fixed analyzer, or a calibrated pulse before a fixed detector basis. For a single context the two sweeps are related by moving the group action from the projectors to the state inside the Born bracket, and the calibration requirements apply verbatim to the preparation knob. An *uncalibrated* state sweep, by contrast, discriminates nothing: by Proposition 1 it covers the entire simplex. At the multi-context level the symmetry is only partial. The gluing problem of Sec. IV is generated by atoms shared between measurement contexts, at one fixed state; the nontrivial state-side analogue is instead preparation noncontextuality in the sense of Spekkens, in which distinct convex decompositions of the same mixed state must receive one ontological representation. The simplex-embedding tests [4, 5] treat both sides symmetrically.

VII. CONCLUSION

The moral is not that quantum probability is secretly classical, nor that a single probability vector contains enough information to reveal its origin. The moral is more local: one must keep track of what has been thrown away. When the generating geometry is forgotten, a single context leaves only a shadow. When the physical knob is retained, the shadow begins to move, and its calibrated, symmetry-constrained motion can distinguish

specified affine, Malus-type, softmax, and threshold response models.

A calibrated apparatus sweep nevertheless distinguishes only model classes whose physical action, covariance, and causal restrictions have been specified. Reparameterizing a plotted scalar is not physically harmless unless the reparameterization intertwines the independently calibrated group action. Conversely, one observed response curve does not by itself exclude arbitrary classical stochastic models or establish a Bell- or Kochen–Specker-type impossibility theorem.

Two weakly pasted shadows still need not betray anything nonclassical. If two contexts agree on their shared atoms, the remaining mass can be coupled by an ordinary joint distribution. The obstruction appears only when enough local shadows are required to hang together as one global object. The question then becomes a linear feasibility problem: either there is a classical joint model or simplex embedding, or Farkas’ lemma supplies a separating inequality. The elastic-band models provide a laboratory parable for this distinction: with adaptive realignment, context communication, or one extended nonseparable object, they can produce quantum-looking pairwise shadows, but not one Bell-local valuation glued

across all four settings.

Thus the staged picture is: shadow, calibrated knob, weak pasting, global web. The frozen shadow hides classical, quantum, and exotic origins. The knob records how the shadow changes under symmetry-constrained variation. Weak pastings remain extendable. Rich webs can fail to extend. Finally, the pentagon half-weight reminds us that admissibility and atom-consistency are not the same as classical embeddability, Hilbert-space realizability, or a physical response law: the same static constraints can be obeyed by very different knobs.

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